Input: Natural-Language Requirement (SRS)

GATE 1: REPRESENTATION SUITABILITY (RSS)

Select candidate representation
(from the 30-representation catalog)

Evaluate suitability metrics M_1–M_{10}
concern alignment, abstraction fit,
Occam's razor, LLM feasibility, ...

Calculate Representational Suitability Score
$\mathrm{RSS} = \sum_i w_i \cdot M_i$

RSS $\geq$ threshold?

No → Reject & re-select representation

Yes

GATE 2: SYNTACTIC FORM VALIDITY (SFV)

Generate test cases in the approved representation

Evaluate Syntactic Form Validity
$\mathrm{SFV} = \dfrac{\text{syntactically correct elements}}{\text{total model elements}}$

SFV $\geq$ threshold?

No → Regenerate / fix syntax errors

Yes

GATE 3: FUNCTIONAL SEMANTIC ADEQUACY (FSA)

Evaluate semantic testing coverage
clause, negative-path, oracle, and boundary coverage

Calculate Functional Semantic Adequacy
$\mathrm{FSA} = w_c C_{\text{clause}} + w_n C_{\text{neg}} + w_o A_{\text{oracle}} + w_b C_{\text{bound}} + w_g M_g$

FSA $\geq$ threshold?

No → Regenerate / add missing test scenarios

Yes

Approve test suite: requirement fully tested